

Characterizing Energy Technical Assistance Designs to Support Strategic Program Development: A scoping review protocol

Administrative Information

1. Identification

This is a **Scoping Review**

2. Authors

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3. Roles and Responsibilities

a. Content expertise: Gabriel Mendez-Sanders

b. Systematic review methods: Alexandra Coso Strong

c. Statistical analysis: N/A; Gabriel Mendez-Sanders will lead data charting and qualitative analysis.

d. Information retrieval: Gabriel Mendez-Sanders

Gabriel Mendez-Sanders:

Gabe is leading all stages of the project. This study is a qualitative scoping review, so there is limited statistical analysis. Gabe has content expertise in energy through energy-focused academic research projects, internships, and technical assistance experience that span public, private, and non-profit sectors.

Alexandra Coso Strong:

Dr. Strong is providing project guidance and expertise on review methods, information retrieval, documentation strategy, screening, data charting, and analysis. She has relevant expertise in theoretical foundations and design for complex systems.

Introduction

1. Background

Describe the rationale for the review in the context of what is already known. Provide a description of the problem, condition, or issue that the intervention under review is aiming to address.

(Citations of relevant papers may be provided.)

Energy technical assistance (ETA) programs leverage the expertise of energy experts to deliver educational offerings (e.g., workshops, fact sheets) and project support (e.g., feasibility studies, financial planning) in order to help stakeholders make informed decisions about implementing energy technologies, practices, or processes. This study will generate the first national evidence base of ETA designs and practices using a scoping review (ScR) to produce evidence maps detailing the ETA landscape. These evidence maps will provide a central reference to support funding entities in identifying the types and target audiences of ETA needed, providers in continuously refining and adopting new practices, and program managers in developing evaluation standards.

Background and Rationale

Unprecedented growth in U.S. energy demand is increasing the need to deploy decentralized and distributed energy resources (DERs) to support electric grid reliability [1, 2]. Because many DERs are implemented by stakeholders with limited engineering or energy expertise, successful deployment depends on stakeholders learning engineering concepts through ETA to make informed decisions about system design, construction, and maintenance. The importance of ETA has been recognized by the U.S. Government Accountability Office, which concluded that effective ETA designs and educational practices will be vital to helping multiple stakeholder groups meet rising energy needs and sustain project success after deployment [3]. ETA is widely used by the U.S. Department of Energy (DOE), state energy agencies, and NGOs to translate large-scale goals into local initiatives [5–6]. DOE alone supports dozens of ETA programs with annual budgets totalling in the hundreds of millions of dollars [7, 8]; for example, in 2023 DOE allocated \$530 million to ETA on zero-energy building codes, and the 2025 President’s Budget dedicated \$95 million to ETA to tribal nations [9, 10]. Despite its importance, ETA designs and evaluation standards remain fragmented across locations and technologies, and recent program evaluations recommend tailoring program design to better meet stakeholder needs [11, 12].

Historically, ETA has evolved in response to shifting policy and market needs (e.g., 1970s energy crisis) rather than through a unified framework or organizational strategy [13–15]. As a result, limited knowledge exists about which program designs and practices are used, in what contexts, and why [11, 16]. Studying ETA designs (i.e., program components and structure) and practices (i.e., recurring activities and their delivery strategies) can establish a national understanding of existing approaches, evidence gaps, and opportunities for strategic program development. This study will visually and conceptually diagram the goals, practices, delivery strategies, and contexts of U.S. ETA programs using a ScR. This approach is particularly appropriate for synthesizing the heterogeneous ETA literature, as the term has been applied across

many contexts without a unifying definition, and much of the evidence exists in grey literature (e.g., program reports, webpages).

References:

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Methods

1. Research Question

Primary Research Question:

What similarities and differences exist in U.S. ETA goals, components, and delivery strategies across contexts (i.e., ETA provider, ETA receiver, geographic location, technology)?

Sub-questions:

- **SRQ 1:** What organizations/entities provide ETA and to whom are they providing it?
- **SRQ 2:** What are the different energy technologies and processes that ETA supports?
- **SRQ 3:** How do ETA programs describe their goals and objectives?
- **SRQ 4:** What practices, activities, and resources (i.e., *tools*) do programs use for ETA and how are they delivered?
- **SRQ 5:** What are the *rules* of participation in ETA programs?
- **SRQ 6:** To what extent and in what ways do U.S. ETA programs explicitly or implicitly describe education and learning in their *reported* goals, design, and delivery strategies?

2. Objectives

a. Setting:

The energy sector in the U.S.

b. Population(s) of interest

ETA programs that are based in the U.S. and targeted towards U.S. stakeholder groups.

c. Phenomenon of interest

Documents or articles that:

- Describe ETA program practices, rules, goals, and/or recruitment efforts.
- Describe ETA frameworks or designs.
- Describe ETA program evaluation, evaluation strategies, or outcomes.
- Are published in English
- Are published in or after the year 2000

d. Comparison

N/A; No comparator is chosen, which is in line with the exploratory nature of a scoping review.

e. Outcome(s) of interest (For more detail, see Section 9 of this protocol, ‘Data Charting’):

- Organizations/entities providing ETA (SRQ1)
- Organizations/entities receiving ETA (SRQ1)
- Types of ETA (i.e., technologies and processes it supports) (SRQ2)
- Program goals and objectives (SRQ3)
- Frequency of terms used for goals and objectives (SRQ3)
- Program practices, activities, and resources (i.e., tools) (SRQ4)
- Delivery strategies of program activities (e.g., in-person vs. virtual, synchronous vs. asynchronous, on-site vs. public meeting, individual vs. group-based) (SRQ4)
- Rules of participation in programs (SRQ5)
- Frequency of terms used for rules (SRQ5)
- Similarities and differences in goals, components, and delivery strategies across contexts (Primary RQ)
- Changes in program goals, components, and delivery strategies over time
- Descriptions of education and learning (SRQ6)
- Philosophical underpinnings or frameworks of ET
- Evaluation techniques and measures of success for ETA

f. Study design/publication type of interest

- English-language literature
- 2000–present
- Peer-reviewed publications from bibliographic databases
- Gray literature:
 - Program reports, recruitment documents, lab presentations, brochures, posters, white papers, subcontract reports, marketing material, and fact sheets from research and government repositories
 - Webpages from program websites

3. Definitions

ACCESSING INFORMATION: DEFINITIONS

Bibliographic databases: Organized digital collections that index and catalog published literature like journal articles, books, and reports by providing structured information (e.g., titles, authors, abstracts, keywords) allowing users to search for and identify relevant sources.

Research and government repositories: Digital collections maintained by research institutions and government agencies (e.g., DOE, NREL, NYSERDA) that provide access to gray literature such as program reports, white papers, technical documents, and other non–peer-reviewed materials.

Program websites (ETA): Official webpages created by ETA programs that share information about services, resources, eligibility, and program activities, which may include materials like brochures, fact sheets, and implementation details intended for prospective participants or program reviewers.

ETA: DEFINITIONS

Energy Technical Assistance (ETA): Any expert-driven non-degree educational activity, training, advising, modelling, or technical support to stakeholders (e.g., organizations, communities, individuals) to enable informed decision-making, skill development, or project implementation for energy technologies, practices, or processes.

Under this definition, a state-funded program that offers workshops and individualized support to school districts installing solar and battery systems would be included as ETA. In contrast, a private engineering firm hired solely to design a system without engaging in stakeholder training or education would not be considered ETA. In general, payment of services would not exclude a practice from being considered ETA. Unlike with consulting, ETA providers will never make decisions for their participants, but rather only support decision making with knowledge, skills, and data. Instead, services are not considered ETA if they are purely transactional or outsourced work/decision-making without aiming to improve the participant's ability to perform future energy-related activities. However, during the scoping review, any programs that explicitly use the term "ETA" will be initially considered.

Technology (energy): Physical systems or equipment related to energy generation, storage, distribution, or efficiency (e.g., solar PV systems, battery storage, heat pumps, smart grid technologies).

Processes (energy-related): Non-physical procedures, workflows, or decision-making activities that support energy planning, implementation, or management (e.g., energy auditing, financial modeling, procurement, policy compliance, regulation development, permitting, project planning, standards).

Contexts: The characteristics and circumstances that define the setting in which an ETA program operates, seeks to support, and/or is implemented. Contexts may include, but are not limited to, participant type, provider organization, geographic location, energy technology type, energy process type, sector, project stage, funding source, or regulatory environment.

ETA PROGRAM: DEFINITIONS

Program Goals: Broad, high-level statements that describe the intended purpose or desired long-term impact of an ETA program.

Program Objectives: Specific, measurable, or time-bound aims that operationalize program goals, often describing what the program seeks to accomplish through its activities.

Practices: Recurring or structured approaches used by ETA programs to deliver support, including pedagogical, technical, or engagement methods (e.g., coaching, needs assessments, peer learning programs, iterative feedback).

Activities: Discrete actions or events carried out as part of a program's implementation, such as workshops, webinars, site visits, or one-on-one consultations.

Resources: Documents, digital programs, databases, or calculators that ETA practitioners give, lend, or grant access to participants and sometimes train them in to support them in pursuing their energy projects/object(ive)s.

Delivery strategies: The modes and formats through which ETA is provided, including how, where, and in what setting activities occur (e.g., in-person vs. virtual, synchronous vs. asynchronous, individual vs. group-based).

Designs (program designs): The overall structure and organization of an ETA program, including how its components (e.g., goals, activities, resources) are sequenced to achieve desired outcomes.

Evaluation: Systematic processes used to assess the effectiveness, quality, or outcomes of an ETA program, which may include formal or informal methods such as surveys, performance metrics, interviews, and/or impact assessments.

ETA Professionals: Individuals or groups of individuals involved in funding, designing, delivering, managing, or evaluating ETA programs.

ETA Providers/Facilitators: Individuals or organizations that directly coordinate or deliver technical assistance to participants through educational activities, technical support, and other program services.

ETA Participants: Individuals, organizations, communities, or other stakeholder groups that receive or engage with ETA to support energy-related planning, policy development, decision-making, implementation, operation, and/or maintenance.

Provider Learning Orientation: A provider's collection of beliefs about the role of learning in ETA, including what participants should learn, how learning occurs, what supports learning, and the provider's role in facilitating learning.

CULTURAL-HISTORICAL ACTIVITY THEORY: DEFINITIONS

Activity theory encompasses a wide variety of social-science theories that seek to describe shared human activities as a socially situated system operating to achieve an object. The concept of human action being mediated by social interactions and cultural tools originates from the work of Russian scholar Lev Vygotsky; his foundational research was later developed into activity theory by Alexei Leontiev and expanded into CHAT by Yrjö Engeström [7, 8, 32, 33].

CHAT describes how an individual (*subject*) engages in a collective activity within a *community* to pursue some goal (*object*) by using *tools*, distributing tasks through power relations (*division of labor*), and following shared *rules*. These *six* components each have relationships to one another, which taken as a whole is an activity system. The components of CHAT can be summarized as follows:

Subject: The principal individual or group engaged in an activity. In ETA, this typically refers to the primary actor(s) delivering or receiving assistance (e.g., ETA providers such as program staff, or participants representing entities like school districts, municipalities, or community organizations).

Object: The goal toward which the activity is directed. In ETA, the object can be different depending on whether you are focusing on the practitioner or participant as subject. Examples of objects could focus on energy project deployment, enabling effective decision-making, and/or developing processes or infrastructure for future energy projects.

Tools: The resources or mediating artifacts used to act on/towards the object. In ETA, these include both physical and informational support in the form of activities, resources, and practices such as software tools, financial models, peer presentations, guidance documents, and training materials used to facilitate learning and project development.

Community: The group of actors who share an interest in the activity. For a ETA practitioner, this may include the program funding agency, colleagues, the program manager, and others in their professional network. For a ETA participant, their community may include organization members, organizational leadership, their utility company, neighbors, collaborators, contractors, and local government officials.

Rules: The formal and informal norms governing participation in the activity and regulating the subject's actions towards the object. In ETA, these include program eligibility criteria, funding requirements, regulatory constraints, participation expectations, and implicit norms shaping how ETA is delivered and received.

Division of labor: The distribution of roles, responsibilities, authority, access to resources, and rewards among participants. In ETA, this refers to how tasks are allocated between providers (e.g., delivering training, conducting analyses) and

recipients (e.g., implementing projects, making decisions), as well as the roles of partners or supporting organizations.

4. Eligibility Criteria

a. Provide a numbered list of study inclusion criteria

1. Describes, studies, or evaluates one or multiple ETA program(s).
2. Program(s) are based in the U.S.
3. Program(s) are targeted towards only /mostly U.S. stakeholders.
4. Program(s) provide ETA to organizations or individuals planning to engage in energy-related planning, decision-making, deployment, or implementation.
5. Program(s) include an educational, training, advisory or capacity-building component, practice, or stated goal.
6. Provides details of program(s') design, delivery, evaluation, or participant engagement to permit data charting.
7. Uses qualitative, quantitative, mixed-methods, descriptive, or evaluative approaches.
8. Published in English.
9. Published between 2000 and 2026.
10. Is a peer-reviewed article, book chapter, webpage or grey literature source (e.g., Program report, program evaluation, recruitment document, lab presentation, brochure, poster, white paper, subcontract report, marketing material, fact sheet).

Justification of Inclusion Criteria:

1. The phenomenon of interest in this scoping review is ETA, with a focus on designs, practices, delivery strategies, and goals; including documents that describe or evaluate ETA support understanding of this phenomena in addition to peer-reviewed research publications.
2. This narrows the scope to domestic programs; this study seeks to inform the design and evaluation of domestic programs only.
3. This study seeks to inform the design and evaluation of domestic programs only. Some domestic ETA programs are primarily targeted towards international groups or entities; those programs are outside the scope of this project.
- 4.
- 5.
6. FILLLLLL
- 7.
- 8.
9. The year 2000 was selected as the start date to focus on contemporary energy system goals and ETA practices. The early and mid 2000s marked a major shift in federal investments towards DERs, grid modernization, and expanded infrastructure, with the expansion of policies supporting decentralized energy deployment and the American Recovery and Reinvestment Act. These shifts brought a wider range of stakeholders into energy decision-making and increased the need for ETA to support non-expert

actors. These shifts in energy systems became even more dominant during the 2021–2025 Biden administration. Thus, ETA practices since 2000 are more representative of contemporary program designs, learning environments, and implementation challenges. This timeframe is also consistent with other reviews of ETA [4].

10. ETA programs are frequently documented outside traditional peer-reviewed literature, particularly by government agencies, national laboratories, utilities, and nonprofit organizations. Grey literature is rich in detail on program design, recruitment, and practices which is necessary for this study's focus, complimenting the controlled research available in peer-reviewed articles and book chapters.

b. Provide a numbered list of study exclusion criteria

1. Focuses solely on K–12, undergraduate, or formal degree-based education.
2. Focused exclusively on energy workforce development within formal degree programs.
3. Describes energy technologies or policies without a ETA component.
4. Program(s) based outside the U.S.
5. Program(s) targeted towards only /mostly non-U.S. stakeholders (e.g., USAID).
6. Opinion pieces or editorials.
7. Lacks details of program design, delivery, evaluation, or participant engagement.
8. Conference abstracts without accessible full-text documentation.
9. Theoretical discussions of adult learning without connection to ETA programs.
10. Program(s) that are not about energy, energy efficiency, energy management, or a related energy topic.
11. Program(s) that focus on climate change, emissions, or environmental impact but not specifically energy and energy usage.
12. Program(s) that focus on healthcare, medical, occupational safety or biomedical applications where energy is related to or used for diagnosis, treatment, or health risk assessment.
13. Discusses ETA or seeks to inform future ETA efforts without describing, studying, or evaluating existing programs.
14. Full-length books.
15. Patents.
16. Software files.

Justification of Exclusion Criteria:

1. These formal educational settings do not encapsulate the non-degree educational activities and programming of ETA delivered to adult participants developing energy projects.
2. Even with a workforce development focus, these formal educational settings do not encapsulate the non-degree educational activities and programming of ETA delivered to adult participants developing energy projects. ETA is not defined by workforce development, but rather by decision-support.
3. Outside the scope of this study which focuses specifically on ETA.
4. Outside the scope of study which has a domestic-focus.

5. Outside the scope of study which has a domestic-focus.
6. Outside the scope of phase 1 of this study, which seeks to document reported ETA designs and practices, not opinions on the energy industry or ETA.
7. Many documents may peripherally mention ETA without substantive description of a program, its design, or its activities. These types of documents will not contribute useful information based on the desired outcomes of this work.
8. Without an accessible full-text document, the work cannot be properly screened or analyzed.
9. While adult learning is relevant to the educational context of ETA, all documents must center ETA as the focus to be incorporated into this study.
10. This review does not seek to integrate technical assistance work on other topics outside of energy; all documents must focus on ETA.
11. This review does not seek to integrate technical assistance work on other topics outside of energy, even those which offer technical assistance on non-energy environmental topics are outside the scope; all documents must focus on ETA.
12. ETA work included must focus on technologies and processes focused on generating, transmitting, distributing, and/or storing energy/electricity. Topics of health and occupational safety are not seeking to develop energy projects or processes as in the described focus of this study.
13. Many documents may peripherally mention ETA without substantive description of a program, its design, or its activities. These types of documents will not contribute useful information based on the desired outcomes of this work.
14. Due to constraints on time of the researcher and the relative low prevalence of full-length books on the topic of ETA, full-length books will not be included.
15. The research team is not aware of any relevant patents on ETA, nor does it appear that intellectual property of ETA practice is a prevalent issue or topic.
16. This work is not interested in analyzing digital tools used by ETA practitioners; rather the focus is on the sociological systems, activities, and interpersonal interactions within ETA. As such, software files are not included in this scoping review.

5. Information Sources

a. Peer-Reviewed Literature Databases

Peer-reviewed literature will be identified through systematic searches of the following multidisciplinary, engineering, and education-focused databases:

- Education Resources Information Center (ERIC)
- Engineering Village (Compendex)
- IEEE Xplore
- Scopus
- Web of Science

Date coverage: Searches will include all available records in each database up to the date of search execution (to be documented in the final PRISMA-ScR flow diagram).

b. Energy and Government/Technical Repositories (Grey Literature)

Grey literature will be systematically searched from institutional and government repositories to compliment peer-reviewed articles with details of ETA designs, practices, outreach, and reporting:

- National Laboratory of the Rockies (NLR) Publications Database
- Office of Scientific and Technical Information, U.S. DOE (OSTI.gov)
- Energy Information Administration (EIA) reports and technical briefs
- National Academies Press (NAP) reports
- EPA and other U.S. federal agency technical reports (where relevant to energy systems and education/training programs)

c. Institutional Websites and Programmatic Materials (Web-based Grey Literature)

In addition to bibliographic databases and grey literature repositories, targeted searches of institutional and organizational websites will be conducted to capture program evidence on models, practices, goals, and participant details that may not be described in academic databases.

Searches will focus specifically on organizations that design, fund, or deliver ETA, including:

- U.S. federal energy agencies and offices (e.g., DOE program offices and ETA initiatives)
- National laboratories involved in energy systems ETA and deployment support (e.g., NLR)
- State and regional energy offices and clean energy program administrators
- Utility-supported or utility-adjacent ETA programs
- Nonprofit and NGO organizations providing energy transition, decarbonization, or efficiency ETA support
- Public-private partnerships and workforce/implementation intermediaries supporting energy system deployment

Examples of relevant programmatic materials include:

- Program websites describing ETA services and eligibility
- ETA toolkits and implementation guides
- Project summaries and case studies from ETA
- Publicly available reports, fact sheets, and program documentation
- Training or support materials associated with ETA delivery

d. Guidance Resources for Search Strategy Development

Search strategy design and grey literature identification were informed by the following resources:

- Cornell's Evidence Synthesis webpage and consultants
- PRISMA-ScR guidelines
- Open Science Framework (OSF) scoping review templates
- Additional ScR reference papers included in the References section

e. Contact with Authors and Experts

Where necessary, corresponding authors of key studies or reports may be contacted to:

- Clarify methodological details
- Request missing data or unpublished appendices
- Identify additional relevant or ongoing studies (including working papers or in-progress evaluations of ETA programs in energy education contexts)

This will be done minimally and only if absolutely necessary.

6. Search Strategy

Below is the full search strategy used for the bibliographic database Scopus:

TITLE-ABS-KEY ("technical assistance"
AND ("energy" OR "energy efficiency" OR "energy storage system*" OR "renewable energy" OR "clean energy" OR "distributed energy resource*" OR "decentralized energy" OR "DER" OR "energy system*" OR "energy sector*" OR "electrification" OR "building energy" OR "energy management" OR "solar")
AND ("goal*" OR "component*" OR "implementation*" OR "delivery" OR "educ*" OR "train*" OR "learn*" OR "workshop*" OR "curricul*" OR "pedagogy" OR "andragogy" OR "practice*" OR "framework*" OR "model*" OR "approach*" OR "outcome*" OR "strateg*" OR "evaluat*"))

Filters applied:

- Language: English
- Publication year: 2000–present
- Affiliation country = United States or Undefined

All search strategies were adapted from this initial version to fit the interface of the corresponding database or search hub and posted alongside the protocol on the [OSF at this link](#).

Study Records

7. Study Records Management

a. Data Management

Describe the mechanisms used to manage records and data throughout the review (e.g., citation management software).

- This scoping review's searching, screening, full text reviewing, and process of extraction will be managed using Covidence for all data gathered from bibliographic databases. Due to the complexity of importing grey literature into Covidence, grey literature will be screened, reviewed, and extracted in a parallel manner using a comprehensive excel spreadsheet.
- All search results will be saved in Zotero for maintaining local access to DOI's, URLs, and PDFs.
- The full protocol will be pre-registered on [the OSF at this hyperlink](#).
- The full search strategy for all databases will be documented and published on OSF for each database and search hub in separate .txt files to ensure methodological transparency and reproducibility.

b. Selection Process

Describe the process used for selecting studies (e.g., two independent reviewers) through each phase of the review:

- Screening
- Eligibility
- Inclusion in data charting/synthesis

[FILL HERE](#)

c. Data Collection Process

i. *Systematic reviews:*

Describe methods for extracting data from reports (e.g., piloted forms, independent extraction, duplicate extraction), and processes for obtaining or confirming data from investigators.

ii. *Scoping reviews/systematic maps:*

Describe methods for charting data from included sources (e.g., calibrated or tested forms), whether charting is independent or duplicated, and processes for obtaining or confirming data.

[FILL](#)

8. Data Synthesis / Charting

Scoping reviews / Evidence gap maps

i. Describe the planned approach for summarizing main results, including:

- Concepts
- Themes
- Types of evidence available

Explain how summaries will be linked to review questions and objectives, and discuss relevance to key groups.

Data will be coded and analyzed using a staged, hybrid deductive–inductive approach aligned with the broad-to-specific structure of the scoping review. In Stage a, coding will remain intentionally descriptive and non-theoretical, focusing on charting core characteristics of ETA programs, including program goals, delivery formats, participant types, and implementation contexts. These codes will undergo the following stages of descriptive synthesis:

1. Spreadsheet tabulation
2. Organization and grouping
3. Narrative description
4. Visual mapping

The resulting evidence maps will map the flow of ETA from provider to participant (see Figure 4 for example), the frequency of terms used to describe program goals (see Figure 5 for example), and the types of frameworks/strategies used for program design.

In Stage b, inductive coding will be used to identify patterns in how learning and knowledge transfer are embedded within ETA activities. This includes both explicit references to education or training and more implicit instructional elements (e.g., factsheets, workshops, peer exchange programming). This phase will capture how education, training, and advisory support are described in ETA practice using bubble charts for key terms and thematic analysis to group practices without imposing predefined learning frameworks. Gaps in understanding of ETA practice will be documented to inform data collection and interview protocols for the multiple case study.

Only after these descriptive and inductive analyses are complete will a more structured, theory-informed coding be applied for Stage c. At this stage, observed patterns in learning-related practices will be interpreted through the lens of andragogical principles and CHAT to examine how adult learning is supported within different ETA designs. This final phase of coding will support the development of a preliminary typology of ETA practices. This typology will group programs based on their goals and how they structure the activity system (i.e., tools, rules, distribution of labor) to deliver educational support. Subsequent sampling for the multiple case study will aim to capture programs of various types from the typology, with outreach to specific exemplar programs identified from websites and program reports in the scoping review.

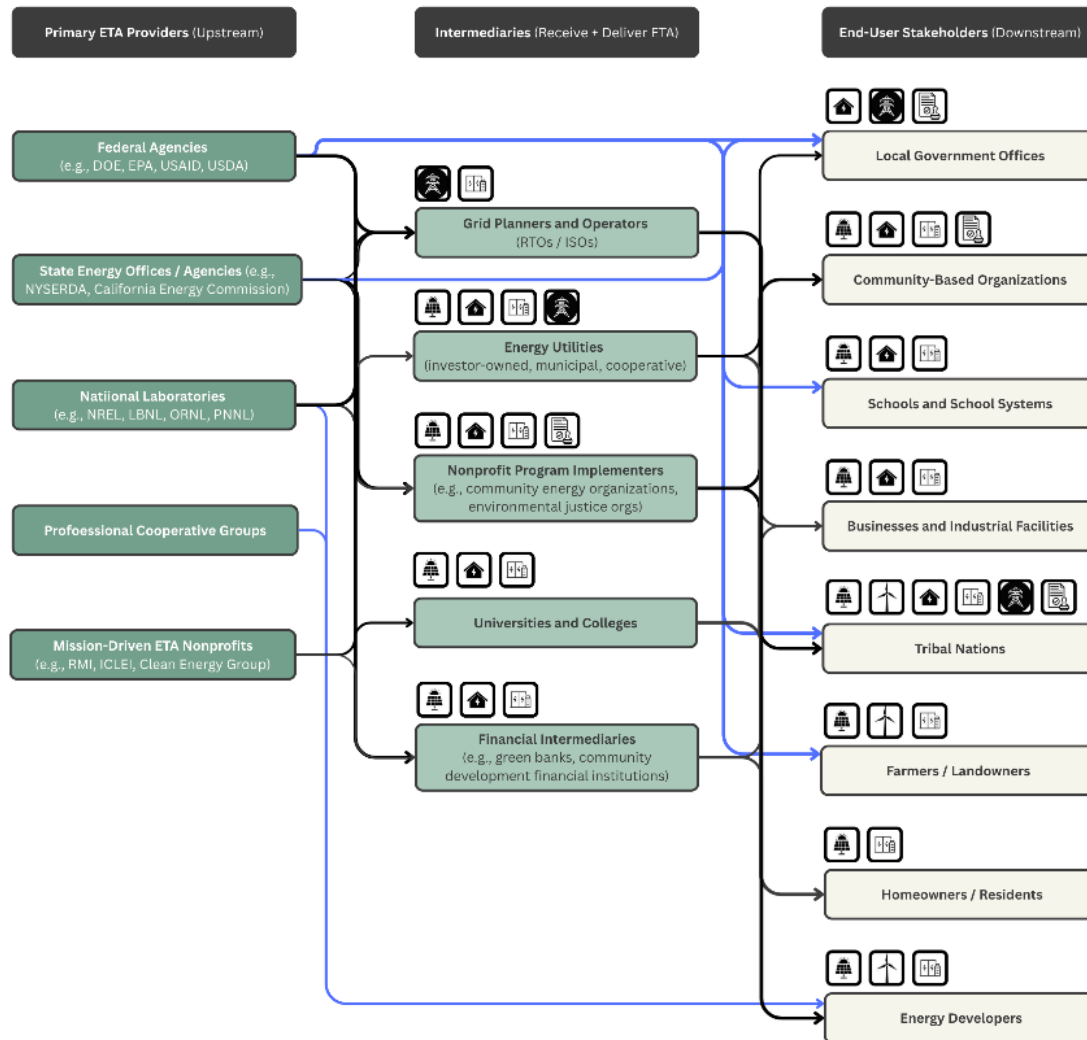
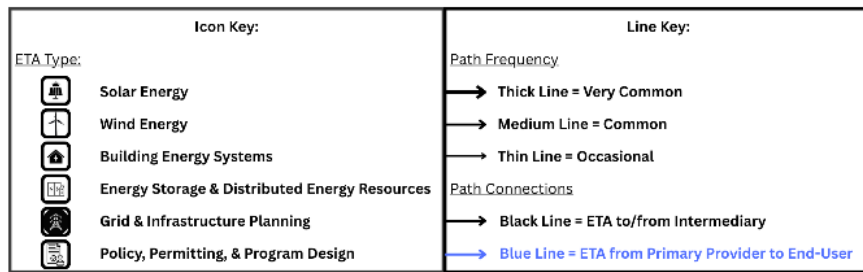


Figure 4. Example ETA provider-participant flow diagram.

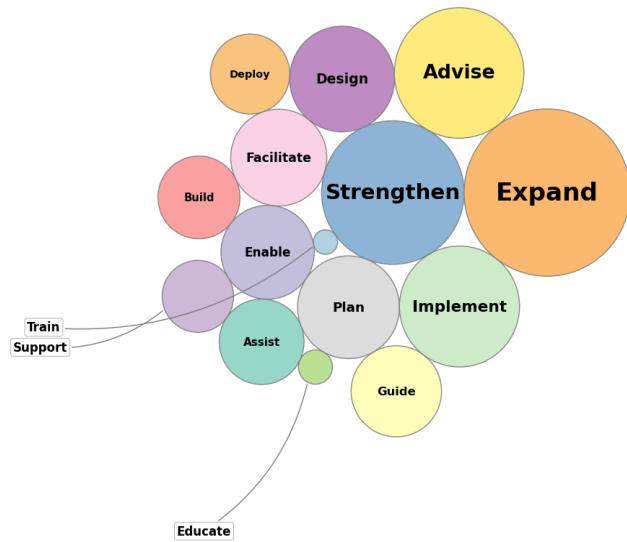


Figure 5. Example ETA program goals verb prevalence bubble chart. *Note:* Bubble size will be proportional to the proportion of identified ETA programs that use that verb in their reported goals.

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Appendix A: Critical Appraisal and Risk of Bias Assessment Resources

1. **CASP Checklists**

Eight critical appraisal tools covering:

- Systematic reviews
- Randomised controlled trials
- Cohort studies
- Case-control studies
- Economic evaluations
- Diagnostic studies
- Qualitative studies
- Clinical prediction rules

<https://casp-uk.net/casp-tools-checklists/>

2. **AMSTAR Checklist**

For evaluation of literature reviews:

https://amstar.ca/Amstar_Checklist.php

3. **Cochrane Risk of Bias Tool (RoB 2.0)**

For randomized controlled trials:

<https://sites.google.com/site/riskofbiastool/welcome/rob-2-0-tool>

4. **Cornell Evidence Synthesis Service LibGuide**

<https://guides.library.cornell.edu/evidence-synthesis/bias>